12-2 Programming Basics [1] (Python)

Point!

Variables and Sequential Structure

(1) ([1]**print**) : Displays the character string or value inside the parentheses ( ).
The character string is enclosed with half-width single quotation marks (') or double quotation marks (").
*Note that numerical values should not be enclosed using half-width quotation marks.

01	`print('Hello Khalid')`	Displays "`Hello Khalid`".
02	`print(2023)`	Displays "`2023`".

Execution result

```
Hello Khalid
2023
```

(2) ([2]**Variable**) : Like a box to store data. Data can be stored in and retrieved from variables.
<Example> A program to display the character string "`Cairo`"

01	`city = 'Cairo'`	Assigns "`Cairo`" to the variable "`city`".
02	`print(city)`	Displays the value for the variable "`city`".

Execution result

```
Cairo
```

* In programming, the "=" symbol does not mean "equal." Rather, it means "([3]**assign the right side to the left side**)."

(3) ([4]**Assignment operator**) : Something such as "=" used to assign a value to a variable.

(4) ([5]**Arithmetic operator**) : A symbol such as "+" and "-" used in calculations.

Arithmetic operator	Meaning
`a + b`	([6]**Addition**)
`a - b`	([7]**Subtraction**)
`a * b`	([8]**Multiplication**)
`a / b`	([9]**Division**)
`a // b`	([10]**Quotient**)
`a % b`	([11]**Remainder when a is divided by b**)
`a ** b`	([12]**a to the power of b**)

<Example> A program that performs four basic arithmetic operations

01	`a = 5`	5 is assigned to variable a.
02	`b = 3`	3 is assigned to variable b.
03	`print(a + b)`	Displays a + b.
04	`print(a - b)`	Displays a - b.
05	`print(a * b)`	Displays a × b.
06	`print(a / b)`	Displays a ÷ b.
07	`print(a // b)`	Displays the quotient when a is divided by b.
08	`print(a % b)`	Displays the remainder when a is divided by b.
09	`print(a ** b)`	Displays a to the power of b.

Execution result

```
8
2
15
1.66666
1
2
125
```

Warm Up

Answer the following questions.

(1) Choose the program that displays "HelloWorld!" from the options **A** to **D** below, and answer using the letter.

A
```
HelloWorld!
```

B
```
'HelloWorld!'
```

C
```
print(HelloWorld!)
```

D
```
print('HelloWorld!')
```

(2) Choose the program that displays "Mr. Suzuki" from the options **A** to **D** below, and answer using the letter.

A
```
print(name)
name = 'Mr. Suzuki'
```

B
```
print(name)
name == 'Mr. Suzuki'
```

C
```
name = 'Mr. Suzuki'
print(name)
```

D
```
name == 'Mr. Suzuki'
print(name)
```

(3) When the following program is executed, give the results for items [1] to [5].

```
a = 6
print(a + 4)     .....[1]
print(a - 1)     .....[2]
print(a * 5)     .....[3]
print(a / 2)     .....[4]
print(a ** 3)    .....[5]
```

Explanation

(1) The `print` function is used to display character strings or values, and the character string "`HelloWorld`"! is enclosed with either ' or ". Therefore, **D**

(2) "`=`" is used to assign a value to a variable. In addition, a variable cannot be used before it is defined. Therefore, **C**

(3) Since the value 6 is assigned to the variable "`a`", [1] is 6 + 4, which equals 10; [2] is 6 - 1, which equals 5; [3] is 6 × 5, which equals 30; [4] is 6 ÷ 2, which equals 3; and [5] is 6 to the power of 3, which equals 216.
Therefore, the answers are [1] `10`, [2] `5`, [3] `30`, [4] `3`, [5] `216`

Try

Answer the following questions.

(1) Choose the program that displays "Hello" from the options **A** to **D** below, and answer using the letter.

A
```
Hello
```

B
```
'Hello'
```

C
```
print(Hello)
```

D
```
print('Hello')
```

(2) Choose the program that displays "Correct" from the options **A** to **D** below, and answer using the letter.

A
```
print(result)
result = 'Correct'
```

B
```
print(result)
result == 'Correct'
```

C
```
result = 'Correct'
print(result)
```

D
```
result == 'Correct'
print(result)
```

(3) Choose the program that displays "2" from the options **A** to **D** below, and answer using the letter.

A
```
a = 20
print(a % 3)
```

B
```
print(a % 3)
a = 20
```

C
```
a = 20
print(a * 3)
```

D
```
print(a * 3)
a = 20
```

(4) Choose the program that displays "81" from the options **A** to **D** below, and answer using the letter.

A
```
b == 9
print(b // 2)
```

B
```
b = 9
print(b // 2)
```

C
```
b == 3
print(b ** 4)
```

D
```
b = 3
print(b ** 4)
```

(5) When the following program is executed, give the results for items [1] to [5].

```
c = 5
print(c + 3)     ……[1]
print(c - 2)     ……[2]
print(c * 4)     ……[3]
print(c / 2)     ……[4]
print(c ** 2)    ……[5]
```

(6) What is the term for a symbol like "=" used to assign a value to a variable? Choose one from the options **A** to **D** below, and answer using the letter.

A Arithmetic operator
B Comparison operator
C Assignment operator
D Logical operator

Exercise

1 Answer the following questions.

(1) Choose the program that displays "`Nice to meet you`" from the options **A** to **D** below, and answer using the letter.

A
```
print('Nice to meet you')
```

B
```
print(Nice to meet you)
```

C
```
'Nice to meet you'
```

D
```
Nice to meet you
```

(2) Choose the program that displays "`17 years old`" from the options **A** to **D** below, and answer using the letter.

A
```
print(age)
age == '17 years old'
```

B
```
print(age)
age = '17 years old'
```

C
```
age == '17 years old'
print(age)
```

D
```
age = '17 years old'
print(age)
```

(3) Choose the program that displays "`15`" from the options **A** to **D** below, and answer using the letter.

A
```
a = 5
print(a % 3)
```

B
```
print(a % 3)
a = 5
```

C
```
a = 5
print(a * 3)
```

D
```
print(a * 3)
a = 5
```

(4) Choose the program that displays "`2`" from the options **A** to **D** below, and answer using the letter.

A
```
b == 8
print(b // 3)
```

B
```
b = 8
print(b // 3)
```

C
```
b == 8
print(b ** 3)
```

D
```
b = 8
print(b ** 3)
```

(5) When the following program is executed, give the results for items [1] to [5].

```
c = 9
print(c + 2)     ……[1]
print(c - 4)     ……[2]
print(c * 2)     ……[3]
print(c // 2)    ……[4]
print(c % 2)     ……[5]
```

(6) What is the term for symbols such as "`+`" and "`-`" that are used in calculations? Choose one from the options **A** to **D** below, and answer using the letter.

A Comparison operator
B Logical operator
C Assignment operator
D Arithmetic operator